

Cost and Input Demand Properties: Summary

In what follows let Y be a single-output, free-disposal technology represented by an increasing and continuous production function, $f : \mathbb{R}_+^n \rightarrow \mathbb{R}$ satisfying $f(0) = 0$, where $n \leq L - 1$; in particular,

$$Y = \{(q, -z) \in \mathbb{R}^{n+1} \mid q \in \mathbb{R}_+, z \in \mathbb{R}_+, \text{ and } q \leq f(z)\}.$$

For $(w, q) \in \mathbb{R}_{++}^n \times \mathbb{R}_+$, define

$$c(w, q) = \min_{z \in \mathbb{R}_+^n} w \cdot z \tag{1}$$

subject to the constraint that $f(z) \geq q$. Clearly, for each $w \gg 0$, $c(w, \cdot)$ is strictly increasing. (Proof?)

Theorem 1. (*Properties of $c(\cdot, q)$.*) For every $q \geq 0$, the function $c(\cdot, q)$ on $\mathbb{R}_{++}^n$ is

- nondecreasing;
- homogeneous of degree 1; and
- concave. □

Let $z^c(w, q)$ be the set of solutions to (1). If $c(\cdot, q)$ is C^1 , then $\partial c(w, q)/\partial w_i = z_i^c(w, q)$ (Shepard's Lemma), implying $z^c(\cdot, q)$ is single-valued. The next result follows from Theorem 1 and Shepard's Lemma.

Corollary 1. (*Properties of $z^c(\cdot, q)$.*) For every $q \geq 0$

- $z^c(\cdot, q)$ is homogeneous of degree 0 on $\mathbb{R}_{++}^n$;
- If $c(\cdot, q)$ is C^2 (implying $z^c(\cdot, q)$ is single-valued and C^1), then the matrix

$$\begin{bmatrix} \partial z_1^c / \partial w_1 & \dots & \partial z_1^c / \partial w_L \\ \vdots & & \vdots \\ \partial z_L^c / \partial w_1 & \dots & \partial z_L^c / \partial w_L \end{bmatrix}$$

is a symmetric, negative semidefinite matrix for all $w \gg 0$.

Corollary 2. If $c(\cdot, q)$ is C^2 for every $q \geq 0$ in its domain, then for every $(w, q) \gg$

$$\sum_{j=1}^L \frac{\partial z_i^c}{\partial w_j} w_j = 0 = \sum_{i=1}^L \frac{\partial z_i^c}{\partial w_j} w_i$$

for $i = 1, \dots, L$ and $j = 1, \dots, L$.

Let $z^*(p, w)$ be the set of solutions to $\pi(p, w) = \max_{z \geq 0} [pf(z) - w \cdot z]$, where $(p, w) \gg 0$ and suppose that for every $(p, w) \gg 0$, $z^*(p, w)$ contains exactly one element. Let $q^*(p, w) = f(z^*(p, w))$ be the output supply function. The next Lemma follows from continuity and monotonicity of f .

Lemma 1. *For every $(p, w) \gg 0$, we have that $z^c(w, q^*(p, w)) = z^*(p, w)$ and $\pi(p, w) = \max_{q \geq 0} \{pq - c(w, q)\}$.*

Suppose that $\pi(\cdot)$ is C^2 , so that z^* and z^c are C^1 . By the Lemma we get the following version of the Slutsky equation for input demand: for every $(p, w) \gg 0$

$$\frac{\partial z_i^*}{\partial w_j} = \frac{\partial z_i^c}{\partial w_j} + \frac{\partial z_i^c}{\partial q} \frac{\partial q^*}{\partial w_j}.$$

This equation decomposes the effect of an input price increase into “substitution” and “output” effects. Now set $j = i$. Of course we know that $\partial z_i^*/\partial w_i$ and $\partial z_i^c/\partial w_i$ are each nonpositive. And for $n \geq 2$, we know that both $\partial z_i^c/\partial q$ and $\partial q^*/\partial w_i$ can be either positive or negative.

Puzzle: Why can't there be an output effect opposite in sign to the substitution effect and sufficiently large so that $\partial z_i^*/\partial w_i > 0$?